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Project Management

BOE070

One-day revision notes: definitions, solved examples, code and important questions

Program: B.Tech | **Branches:** ALL | **Semester:** 7 | **Credits:** 3 | **L-T-P:** 3-0-0

Scheme: AKTU B.Tech 4th year, Open Elective-II, 2025-26 onwards

UNIT 1	UNIT 2	UNIT 3	UNIT 4	UNIT 5
Project Concepts and Planning	Project Selection, Scheduling and Risk	Procurement, Quality and Resource Management	Project Control, Evaluation and Close-out	Site Management, Finance and International Projects

Original revision notes written for the AKTU syllabus. Every unit has the same parts:

1. **2-mark definitions**: short answers you can write word for word in Section A.
2. **Core concepts**: frameworks, processes and comparison tables.
3. **Solved examples**: the numericals that appear every year (CPM floats, PERT probability, crashing, breakeven, NPV and payback, decision tree EMV, resource levelling, earned value, replacement analysis).
4. **Formulas / key points to remember**.
5. **Important questions**: the long questions to prepare, with what your answer must contain.

How to use these notes the day before the paper

- **Morning (3 hours):** Unit 1 (project concepts, PMBOK, life cycle, charter, planning, feasibility, SWOT) and Unit 2 (project selection, breakeven, DCF, estimation, CPM, PERT, crashing, risk, decision trees). Unit 2 carries the numericals, so practise them.
- **Afternoon (3 hours):** Unit 3 (vendor selection, JIT, supply chain, quality tools, ISO, TQM, Six Sigma, resource levelling) and Unit 4 (change control, monitoring, earned value, reporting, audit, close-out, shutdown and turnaround projects, replacement analysis).
- **Evening (2 hours):** Unit 5 (site selection and layout, equipment and leasing, safety and environment, project finance, international projects, joint ventures, handover). Then the "Last-hour checklist".
- **In the exam:** draw the network neatly with ES/EF/LS/LF in boxes, write the formula before each calculation, and underline the critical path.

AKTU paper pattern (theory, 70 marks, 3 hours): Section A has 7 short questions of 2 marks, all compulsory (14). Section B has 5 questions of 7 marks, attempt any 3 (21). Section C has 5 questions (one per unit), each with two parts, attempt one part of each, 7 marks each (35). Every unit gets its own Section C question, so do not skip any unit.

UNIT 1 Project Concepts and Planning

✍️ 2-mark definitions

- **Project (PMI):** a temporary endeavour undertaken to create a unique product, service or result. It has a definite start and end.
- **Project management:** the application of knowledge, skills, tools and techniques to project activities to meet the project requirements.
- **PMBOK:** the Project Management Body of Knowledge, the PMI guide that sets out standard processes, knowledge areas and good practices.
- **Triple constraint (iron triangle):** scope, time and cost, with quality in the centre; changing one affects the others.
- **Stakeholder:** any person or organization that is involved in or affected by the project (sponsor, customer, team, suppliers, public).
- **Project sponsor:** the senior person or group that provides resources and support for the project and is accountable for its business success.
- **Project charter:** the document that formally authorizes the project and gives the project manager authority to use resources; it states purpose, objectives, high-level scope, milestones, budget and stakeholders.
- **Scope:** the work that must be done to deliver the product with the required features and functions.
- **Work Breakdown Structure (WBS):** a hierarchical breakdown of the total project work into smaller, manageable deliverables and work packages.
- **Baseline plan:** the approved version of the scope, schedule and cost plan, against which actual performance is measured.
- **Feasibility study:** analysis of whether a project is viable technically, economically, financially, legally and operationally before committing resources.
- **SWOT analysis:** assessing Strengths, Weaknesses (internal), Opportunities and Threats (external).

💡 Core concepts

Characteristics of a project: a definite objective, a life span (temporary), unique, needs a team from many disciplines, involves uncertainty and risk, works within limited resources, has a customer or sponsor, progressive elaboration (details become clearer as the project moves on), and needs change management.

Project vs operations

Feature	Project	Operations
Duration	Temporary	Ongoing
Output	Unique	Repetitive

Feature	Project	Operations
Objective	Achieve a specific goal then end	Sustain the business
Team	Formed for the project	Permanent
Risk	Higher	Lower
Example	Building a metro line	Running the metro daily

Types of projects: by sector (industrial, IT, construction, research, social), by ownership (public, private, PPP, joint venture), by size (small, medium, mega), by urgency (normal, crash, disaster), by location (national, international), by nature (new, expansion, modernization, replacement, diversification).

PMBOK knowledge areas (ten): integration, scope, schedule, cost, quality, resource, communications, risk, procurement and stakeholder management. **Five process groups:** initiating, planning, executing, monitoring and controlling, closing.

Role of the project manager: planner, organizer, integrator (ties all parts together), leader and motivator, communicator, decision maker, negotiator, problem solver, risk manager, and the main link between the team, sponsor and customer. Skills: leadership, communication, technical knowledge, negotiation, time management, team building, conflict resolution.

Project organization structures

Structure	Description	Pros	Cons
Functional	Project done inside a functional department	Expertise, simple	Weak PM, slow, poor customer focus
Pure project (projectized)	Separate self-contained team led by the PM	Full authority, fast, clear focus	Duplication of resources, staff insecurity at the end
Matrix (weak, balanced, strong)	Staff report to both functional head and PM	Uses resources well, balance	Two bosses, conflict

Project life cycle: (1) conception/initiation (idea, feasibility, charter), (2) planning/definition (scope, WBS, schedule, budget), (3) execution/implementation, (4) monitoring and control (along with execution), (5) closure/termination (handover, review). Effort and cost are low at the start, highest during execution, and fall at closure; the ability to influence the project is highest at the start.

Idea generation: sources: market gaps, customer needs, government policies, new technology, import substitution, SWOT of the firm, R&D. Techniques: brainstorming, Delphi, SWOT, environmental scanning, study of trade data, consultants. Screening filters ideas on market, inputs, technology, cost and risk.

Project planning steps

1. Understand customer needs and requirements (voice of the customer).
2. Identify stakeholders and their interests (stakeholder register, power-interest grid).
3. Define scope and objectives (SMART objectives: Specific, Measurable, Achievable, Relevant, Time-bound).
4. Prepare WBS and activity list.
5. Do feasibility analysis.
6. Estimate time, cost and resources; schedule activities.

7. Plan quality, risk, communication and procurement.
8. Create and approve the baseline plan.
9. Form the team and define roles (RACI matrix: Responsible, Accountable, Consulted, Informed).

Types of feasibility: market (is there demand?), technical (can it be built?), financial (can we fund it and does it pay?), economic (social cost-benefit), legal and environmental, operational (will it work in practice?), schedule.

Project team structure: project manager, project team members, functional managers, steering committee, sponsor, project management office (PMO). Team development stages (Tuckman): forming, storming, norming, performing, adjourning.

🧮 Solved examples

Example 1: SWOT for an electric scooter startup.

Strengths	Weaknesses
Battery design patents, young skilled team	Limited capital, no dealer network
Opportunities	**Threats**
FAME subsidies, rising fuel prices, urban demand	Large competitors, battery supply from imports, changing policy

Example 2: WBS for a college fest. Level 1: Fest. Level 2: Planning, Sponsorship, Events, Logistics, Publicity, Closure. Level 3 under Logistics: Venue booking, Stage and sound, Food stalls, Security, Transport. Each level-3 item is a work package with an owner, duration and budget.

Key points to remember

- Project = temporary + unique + specific objective.
- PMBOK: 10 knowledge areas, 5 process groups.
- Structures: functional, projectized, matrix.
- Life cycle: initiation, planning, execution, monitoring and control, closure.
- Charter authorizes; baseline measures; WBS breaks down scope.
- SMART objectives; RACI; Tuckman stages.

⭐ Important questions

1. Define a project. Explain its characteristics and types. Compare project and operations.
2. Explain the role of a project manager and the forms of project organization.
3. Explain the project life cycle with a diagram.
4. What is a project charter? What is the role of a project sponsor?
5. Explain the steps of project planning: customer needs, stakeholders, scope, feasibility, baseline plan, SWOT and team structure.

UNIT 2 Project Selection, Scheduling and Risk

✍️ 2-mark definitions

- **Breakeven point (BEP):** the output level at which total revenue equals total cost, so there is neither profit nor loss.
- **Discounted cash flow (DCF):** valuing future cash flows by discounting them to present value at a required rate of return.
- **Net Present Value (NPV):** the present value of cash inflows minus the present value of outflows; accept if NPV > 0.
- **Internal Rate of Return (IRR):** the discount rate at which NPV = 0; accept if IRR > cost of capital.
- **Payback period:** time needed to recover the initial investment from cash inflows.
- **Profitability index (PI):** PV of inflows / initial investment; accept if PI > 1.
- **Gantt (bar) chart:** a chart showing activities as horizontal bars against a time scale.
- **CPM (Critical Path Method):** a deterministic network technique that finds the longest path (critical path) and floats, with known single activity times.
- **PERT (Program Evaluation and Review Technique):** a probabilistic network technique that uses three time estimates (optimistic, most likely, pessimistic).
- **Critical path:** the longest path through the network; it fixes the project duration, and its activities have zero total float.
- **Total float:** the time an activity can be delayed without delaying the project: LS − ES (or LF − EF).
- **Free float:** the time an activity can be delayed without delaying the early start of any following activity.
- **Crashing:** reducing an activity's duration by adding resources at extra cost.
- **Risk:** an uncertain event that, if it occurs, affects project objectives positively or negatively.
- **Decision tree:** a diagram of decisions, chance events, probabilities and payoffs used to choose the option with the best expected monetary value (EMV).

💡 Core concepts

Project selection methods

Type	Methods
Non-numeric	Sacred cow, operating necessity, competitive necessity, product line extension, comparative benefit
Numeric, non-discounted	Payback period, average rate of return (ARR)
Numeric, discounted	NPV, IRR, profitability index, discounted payback
Scoring models	Weighted scoring of criteria such as profit, risk, strategic fit
Others	Cost-benefit analysis, real options, portfolio analysis

Cost, price and value estimation

- **Cost estimation methods:** analogous (top-down from similar past projects), parametric (using unit rates, e.g. ₹ per sq. m), bottom-up (sum of work packages), three-point estimate.
- **Cost components:** direct (labour, material, equipment), indirect (overheads), contingency reserve (known risks), management reserve (unknown risks).
- **Price:** what the customer pays = cost + profit margin, set by cost-plus, market, competitive bidding or value-based pricing.
- **Value:** the worth to the customer (function/cost); value engineering removes cost without reducing function.

Network rules (activity on arrow and activity on node)

- Each activity is represented by one arrow (AOA) or one node (AON).
- Events are numbered so the tail number is less than the head number (Fulkerson's rule).
- No loops; no dangling activities.
- Dummy activities (dashed, zero time) show logical dependency or keep two activities from sharing the same start and end events.

CPM calculations

- Forward pass: ES of the first activities = 0; EF = ES + t; ES of an activity = maximum EF of its predecessors.
- Backward pass: LF of the last activities = project duration; LS = LF − t; LF of an activity = minimum LS of its successors.
- Total float = LS − ES; free float = ES of successor (minimum) − EF; independent float = ES(successor) − LF(predecessor) − t, taking 0 if negative.

PERT

- $t_e = (a + 4m + b) / 6$; variance $\sigma^2 = ((b - a) / 6)^2$.
- Project duration T_e = sum of t_e on the critical path; project variance = sum of variances on it.
- Probability of finishing by T_s: $Z = (T_s - T_e) / \sigma$, read from the normal table.

CPM vs PERT

CPM	PERT
Deterministic, one time estimate	Probabilistic, three time estimates
Activity-oriented	Event-oriented
Used for repetitive projects (construction)	Used for R&D, new projects
Considers time-cost trade-off (crashing)	Focuses on time and its uncertainty
Critical path from floats	Probability of completion

Crashing (time-cost trade-off)

- Cost slope = (crash cost − normal cost) / (normal time − crash time).
- Steps: find the critical path; crash the critical activity with the lowest cost slope by one unit; recompute paths (new critical paths can appear); continue until required time is reached or crashing costs more than it saves (indirect cost savings), giving the optimum duration.

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- Total project cost = direct cost + indirect cost; the optimum duration minimizes it.

Risk basics: risk management process: identify (brainstorming, checklists, SWOT, Delphi), analyse (qualitative: probability-impact matrix; quantitative: EMV, sensitivity, Monte Carlo), plan responses (avoid, transfer, mitigate, accept for threats; exploit, share, enhance, accept for opportunities), monitor and control (risk register).

Probability concepts used: expected value $E = \Sigma\, p_i\, x_i$; addition rule; multiplication rule for independent events; normal distribution for PERT.

🧮 Solved examples

Example 1: Breakeven. Fixed cost ₹5,00,000; selling price ₹250 per unit; variable cost ₹150 per unit.

- Contribution per unit = 250 − 150 = ₹100.
- BEP (units) = 5,00,000 / 100 = **5,000 units**; BEP (sales) = 5,000 × 250 = ₹12,50,000.
- Profit at 7,000 units = 7,000 × 100 − 5,00,000 = ₹2,00,000.

Example 2: NPV and payback. Investment ₹10,00,000; inflow ₹3,00,000 per year for 5 years; discount rate 10%.

- Annuity factor (10%, 5 years) = 3.7908.
- PV of inflows = 3,00,000 × 3.7908 = ₹11,37,236. **NPV = ₹1,37,236 > 0, accept**.
- PI = 11,37,236 / 10,00,000 = 1.14.
- Payback = 10,00,000 / 3,00,000 = **3.33 years**.

Example 3: CPM and PERT. Times in weeks (a, m, b):

Activity	Predecessor	a	m	b	t_e	σ²
A	–	2	4	6	4	0.444
B	–	2	4	12	5	2.778
C	A	1	2	3	2	0.111
D	A	5	7	9	7	0.444
E	B, C	2	3	10	4	1.778
F	D, E	1	2	3	2	0.111

Forward and backward pass:

Activity	t_e	ES	EF	LS	LF	Total float
A	4	0	4	0	4	0
B	5	0	5	2	7	2
C	2	4	6	5	7	1
D	7	4	11	4	11	0
E	4	6	10	7	11	1
F	2	11	13	11	13	0

- Paths: A-D-F = 13, A-C-E-F = 12, B-E-F = 11. **Critical path A-D-F, T_e = 13 weeks**.
- Variance on critical path = 0.444 + 0.444 + 0.111 = 1.0, so σ = 1.
- Probability of finishing in 14 weeks: Z = (14 − 13) / 1 = 1.0, P = **0.8413 (84%)**. In 12 weeks: Z = −1, P = 0.1587.

Example 4: Crashing the same project. Possible crashing: A by 1 week at ₹2,000; D by up to 2 weeks at ₹1,500 per week; F cannot be crashed.

- To 12 weeks: crash D by 1 week (cheapest on the critical path), cost ₹1,500. Now A-D-F = 12 and A-C-E-F = 12 (two critical paths).
- To 11 weeks: crashing D alone would leave A-C-E-F at 12. Crash A by 1 week (common to both paths), cost ₹2,000. Now all three paths are 11.
- Total extra cost to finish in 11 weeks = **₹3,500**.

Example 5: Decision tree. A firm can build a large plant (cost ₹50 lakh) or a small plant (cost ₹20 lakh). Demand is high with probability 0.6 or low with 0.4. Payoffs (₹ lakh): large plant 120 (high), 40 (low); small plant 60 (high), 40 (low).

- EMV(large) = 0.6 × 120 + 0.4 × 40 − 50 = 72 + 16 − 50 = **38**.
- EMV(small) = 0.6 × 60 + 0.4 × 40 − 20 = 36 + 16 − 20 = **32**.
- Choose the large plant (higher EMV).

Formulas

Item	Formula
BEP (units)	Fixed cost / (price − variable cost)
NPV	Σ CF_t / (1 + r)^t − initial investment
PI	PV of inflows / investment
PERT time	t_e = (a + 4m + b) / 6
PERT variance	((b − a) / 6)²
Probability	Z = (T_s − T_e) / σ
Total float	LS − ES
Cost slope	(C_c − C_n) / (T_n − T_c)
EMV	Σ p_i × payoff_i

⭐ Important questions

1. Explain project selection methods. Explain breakeven analysis and DCF methods (NPV, IRR, PI) with examples.
2. Explain cost, price and value estimation in projects.
3. Draw the network, find the critical path and the floats for given activities (CPM).
4. Explain PERT. Find the probability of completing the project in a given time.
5. What is crashing? Explain the time-cost trade-off with an example.

6. Explain risk management and the use of probability and decision trees.

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UNIT 3 Procurement, Quality and Resource Management

✍️ 2-mark definitions

- **Vendor (supplier) selection:** choosing suppliers on criteria such as quality, price, delivery, capacity, financial stability and service.
- **Just-in-Time (JIT):** a system where materials arrive exactly when needed for production, cutting inventory and waste.
- **Supply chain:** the network of suppliers, manufacturers, warehouses, distributors and retailers that moves materials and products to the customer.
- **Quality:** the degree to which a product meets requirements and is fit for use (Juran: fitness for use; Crosby: conformance to requirements).
- **Quality circle:** a small group of workers from the same area who meet regularly to identify, analyse and solve work-related problems.
- **Quality control (QC):** inspecting and testing outputs to find and fix defects (product-oriented, reactive).
- **Quality assurance (QA):** planned, systematic activities that give confidence that quality requirements will be met (process-oriented, preventive).
- **Cause-and-effect diagram:** the Ishikawa (fishbone) diagram that links a problem to its possible causes grouped under headings (6M: man, machine, method, material, measurement, mother nature/environment).
- **ISO 9001:** the international standard for quality management systems.
- **TQM (Total Quality Management):** an organization-wide approach where everyone continuously improves processes to satisfy customers.
- **Six Sigma:** a data-driven method aiming at no more than 3.4 defects per million opportunities, using DMAIC (Define, Measure, Analyse, Improve, Control).
- **Resource levelling:** adjusting start dates of activities (using float) so that resource demand is smooth and within the limit, which may extend the project.
- **Resource smoothing:** evening out resource use without changing the project end date (only float is used).

💡 Core concepts

Vendor selection process: identify need → search for suppliers → prequalify (questionnaires, visits) → request for quotation/proposal → evaluate (weighted scoring) → negotiate → select and award contract → monitor performance (vendor rating). Contract types: fixed price, cost plus, time and material.

JIT principles: pull system (kanban), small lot sizes, reliable suppliers close by, zero defects, reduced set-up times, continuous improvement (kaizen), elimination of seven wastes (overproduction, waiting, transport, over-processing, inventory, motion, defects). Benefits: less inventory and space, better quality, faster flow. Risks: supply disruption stops work.

Supply chain in projects: procurement planning, sourcing, logistics, inventory, coordination with contractors; aims at right material, right quantity, right time, right place, right cost. Bullwhip effect: small

demand changes amplify upstream.

Quality concepts

- Cost of quality = prevention costs + appraisal costs + internal failure costs + external failure costs.
- Deming's 14 points and the PDCA cycle (Plan-Do-Check-Act).
- Juran's trilogy: quality planning, quality control, quality improvement.
- Crosby: zero defects, quality is free.

Seven basic quality tools: check sheet, histogram, Pareto chart (80/20), cause-and-effect diagram, scatter diagram, control chart, flow chart (or stratification).

QA vs QC

QA	QC
Process-oriented	Product-oriented
Prevents defects	Detects defects
Audits, standards, training	Inspection, testing
Whole team responsible	Inspection team

ISO 9001:2015 principles: customer focus, leadership, engagement of people, process approach, improvement, evidence-based decision making, relationship management. Certification: gap analysis → documentation → implementation → internal audit → management review → certification audit.

TQM elements: customer focus, total employee involvement, process-centred, integrated system, strategic approach, continual improvement, fact-based decisions, communication.

Six Sigma: DMAIC for existing processes, DMADV for new designs; roles: Champion, Master Black Belt, Black Belt, Green Belt. Sigma level 6 = 3.4 DPMO; 3 sigma ≈ 66,807 DPMO.

Resource planning: list resources per activity (labour, equipment, materials), draw a resource histogram (resource vs time from the early start schedule), compare with availability, then smooth or level. Resource-constrained scheduling uses priority rules (least float first, shortest duration first).

Crashing with resources: adding resources to critical activities to cut time (see Unit 2), subject to resource limits.

🧮 Solved examples

Example 1: Weighted vendor rating.

Criterion	Weight	Vendor X score (10)	Vendor Y score (10)
Quality	0.4	8	9
Price	0.3	9	6
Delivery	0.2	7	8
Service	0.1	6	9

- X = 0.4×8 + 0.3×9 + 0.2×7 + 0.1×6 = 3.2 + 2.7 + 1.4 + 0.6 = **7.9**.

- Y = 0.4×9 + 0.3×6 + 0.2×8 + 0.1×9 = 3.6 + 1.8 + 1.6 + 0.9 = **7.9**. A tie: use the most important criterion (quality) to break it, so choose Y, or negotiate price with Y.

Example 2: Resource smoothing. Use the Unit 2 project (A-D-F critical, 13 weeks). Workers needed: A 3, B 2, C 2, D 3, E 2, F 2.

Early-start schedule (A weeks 1–4, B 1–5, C 5–6, D 5–11, E 7–10, F 12–13):

Weeks	Activities running	Workers
1–4	A, B	5
5	B, C, D	7
6	C, D	5
7–10	D, E	5
11	D	3
12–13	F	2

The peak is 7 in week 5. C has 1 week of total float, so start C one week later (weeks 6–7). E must follow C, so E moves to weeks 8–11, using its own 1 week of float, and still ends before F starts in week 12.

Weeks	Activities running	Workers
1–4	A, B	5
5	B, D	5
6–7	C, D	5
8–11	D, E	5
12–13	F	2

The peak falls from 7 to 5 and the project still ends in 13 weeks. This is **smoothing**, because only float was used. If only 4 workers were available, no use of float could meet the limit and the project would have to be extended; that is **levelling**.

Example 3: Pareto analysis. Defects in 200 items: wrong dimension 90, scratches 50, missing parts 30, colour 20, others 10. Cumulative: 45%, 70%, 85%, 95%, 100%. Fixing the first two causes removes 70% of defects, so focus there first.

Key points to remember

- Vendor selection: weighted scoring; contract types.
- JIT: pull, kanban, zero inventory, seven wastes.
- QC detects, QA prevents; PDCA; seven QC tools; cost of quality.
- ISO 9001 seven principles; TQM; Six Sigma 3.4 DPMO, DMAIC.
- Smoothing keeps end date; levelling may extend it.

⭐ Important questions

1. Explain vendor selection and the role of JIT and supply chain management in projects.
2. Explain quality concepts: quality circles, QC, QA, cause-and-effect analysis.
3. Explain ISO 9000, TQM and Six Sigma.
4. Explain resource planning, resource levelling and smoothing with an example.

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UNIT 4 Project Control, Evaluation and Close-out

✍️ 2-mark definitions

- **Change request:** a formal proposal to modify scope, schedule, cost or any baseline document.
- **Change control board (CCB):** the group that reviews, approves or rejects change requests.
- **Project control:** comparing actual performance with the baseline and taking corrective action.
- **Earned value (EV):** the budgeted cost of the work actually completed (BCWP).
- **Planned value (PV):** the budgeted cost of work scheduled to be done by now (BCWS).
- **Actual cost (AC):** the actual cost of work done (ACWP).
- **Project audit:** an independent, systematic review of a project's progress, performance, compliance and management.
- **Project close-out:** formally finishing all activities, handing over deliverables, closing contracts, releasing resources and recording lessons learned.
- **Shutdown (turnaround) project:** a planned stop of a running plant for maintenance, inspection, repair or upgrade, done in a short fixed window.
- **Replacement analysis:** deciding when an old asset should be replaced, by comparing its costs with those of a new one.

💡 Core concepts

Change control process: submit change request → log it → impact analysis (scope, time, cost, quality, risk) → CCB decision → update baselines and plans → communicate → implement → verify. Prevents scope creep (uncontrolled scope growth).

Controlling schedule, resources, cost and quality

- Schedule control: compare actual and planned progress (Gantt, milestone charts), use float, fast tracking (overlap activities) or crashing.
- Cost control: budgets, cost reports, earned value analysis, variance analysis.
- Resource control: monitor usage against plan, reallocate.
- Quality control: inspections, tests, control charts, audits.

Earned value management (EVM)

Measure	Formula	Meaning
Schedule variance	SV = EV − PV	Negative means behind schedule
Cost variance	CV = EV − AC	Negative means over budget
Schedule performance index	SPI = EV / PV	Less than 1 means behind
Cost performance index	CPI = EV / AC	Less than 1 means over budget
Estimate at completion	EAC = BAC / CPI	Expected total cost
Estimate to complete	ETC = EAC − AC	Remaining cost

Measure	Formula	Meaning
Variance at completion	VAC = BAC − EAC	Expected over or under run

Communication and reporting: communication plan (who needs what, when, how), status reports, progress reports, variance reports, dashboards, meetings, milestone reports. Communication channels = n(n − 1) / 2 for n people.

Project audit: types: general audit, detailed audit, technical audit; timing: during the project (to correct) and post-project (to learn). Covers objectives, schedule, budget, quality, risk and management practices. Audit report: introduction, current status, future status, critical management issues, risk assessment, lessons learned.

Project evaluation: ex-ante (before), on-going (during), ex-post (after); criteria: efficiency, effectiveness, impact, relevance, sustainability. Post-implementation review compares benefits with the business case.

Project close-out (termination): types: extinction (completed or killed), addition (becomes a department), integration (resources absorbed back), starvation (budget cut). Activities: confirm deliverables accepted, close contracts, final accounts, release team, archive documents, final report, lessons learned, celebrate.

Maintenance and shutdown projects: preventive, predictive and breakdown maintenance. A shutdown is planned months ahead, has a fixed duration, many contractors working in parallel, strict safety (permits, lockout-tagout), detailed hour-by-hour schedules; any delay causes large production loss.

Plant turnaround: a large planned shutdown of a whole plant (refinery, power plant) every few years for major inspection, overhaul and upgrades. Phases: planning and scoping, pre-shutdown preparation (material and contractor procurement), execution (24×7), start-up, post-turnaround review.

Replacement analysis: replace when the average annual cost (capital cost less salvage, plus cumulative maintenance, divided by years) reaches its minimum; or when the running cost next year exceeds the current minimum average cost. Types: items that deteriorate gradually and items that fail suddenly (group replacement).

🧮 Solved examples

Example 1: Earned value. BAC = ₹10,00,000; at month 4, PV = ₹4,00,000, EV = ₹3,20,000, AC = ₹4,00,000.

- SV = 3,20,000 − 4,00,000 = −₹80,000 (behind schedule); SPI = 0.8.
- CV = 3,20,000 − 4,00,000 = −₹80,000 (over budget); CPI = 0.8.
- EAC = 10,00,000 / 0.8 = ₹12,50,000; ETC = 12,50,000 − 4,00,000 = ₹8,50,000; VAC = −₹2,50,000.

Example 2: Replacement analysis. Machine cost ₹6,000, negligible resale value. Maintenance costs by year: 1,000; 1,200; 1,400; 1,800; 2,300; 2,800.

Year n	Maintenance	Cumulative	Total cost (6,000 + cumulative)	Average per year
1	1,000	1,000	7,000	7,000
2	1,200	2,200	8,200	4,100
3	1,400	3,600	9,600	3,200

Year n	Maintenance	Cumulative	Total cost (6,000 + cumulative)	Average per year
4	1,800	5,400	11,400	2,850
5	2,300	7,700	13,700	2,740
6	2,800	10,500	16,500	2,750

The average cost is lowest (₹2,740) at year 5, so **replace the machine after 5 years**.

Example 3: Communication channels. A team of 10 has 10 × 9 / 2 = 45 channels; adding 5 people gives 15 × 14 / 2 = 105, which is why large teams need a clear communication plan.

Key points to remember

- Change control: request, impact analysis, CCB, baseline update.
- EVM: SV = EV − PV, CV = EV − AC, SPI, CPI, EAC = BAC/CPI.
- Audits during and after; close-out by extinction, addition, integration, starvation.
- Shutdown and turnaround: fixed window, detailed planning, safety.
- Replace at minimum average annual cost.

⭐ Important questions

1. Explain change control and the control of schedule, resources, cost and quality.
2. Explain earned value analysis with an example.
3. Explain project communication, reporting and audit.
4. Explain project evaluation and close-out.
5. Explain maintenance and shutdown projects and plant turnaround.
6. Explain replacement analysis with an example.

UNIT 5 Site Management, Finance and International Projects

✍️ 2-mark definitions

- **Site mapping:** surveying and recording the physical features of a site (boundaries, contours, soil, utilities, access) on drawings.
- **Site layout:** the arrangement of temporary and permanent facilities on site: buildings, stores, roads, cranes, offices, utilities.
- **Site suitability:** how well a location meets project needs: land, access, utilities, labour, markets, environment and laws.
- **Equipment leasing:** using equipment by paying rent for a period instead of buying it; operating lease (short, lessor maintains) or finance lease (long, lessee takes most risks).
- **Project finance:** funding a project mainly on the strength of its own future cash flows, usually through a special purpose vehicle (SPV), with limited or no recourse to the sponsors.
- **Joint venture (JV):** an arrangement where two or more parties pool resources for a specific project, sharing control, risk and profit.
- **Handover:** formal transfer of the completed project, with documents, training and warranties, from the contractor or project team to the owner or operations team.
- **Environmental Impact Assessment (EIA):** a study of the likely environmental effects of a project before approval, with mitigation measures.

💡 Core concepts

Site selection factors: availability and cost of land, access (road, rail, port, airport), raw materials, power, water, labour, markets, climate, soil and topography, environmental clearances, government incentives, community acceptance, safety and disaster risks.

Site preparation: survey and soil testing, clearing and grubbing, demolition, levelling and grading, drainage, temporary roads, fencing, temporary power and water, site offices, stores, labour camps.

Site layout principles: minimize material handling, safe separation of hazardous areas, clear access for emergency vehicles, crane reach, stores near work fronts, keep future expansion space, follow safety and environmental rules.

Equipment selection: factors: suitability for the job, output capacity, cost (owning and operating), availability and standardization, maintenance support, operator skill, site conditions, mobility, flexibility.

Buy vs lease

Buy	Lease
High initial cost	Low initial outlay
Ownership and resale value	No ownership (unless purchase option)
Depreciation tax benefit	Lease rent is tax-deductible
Suits long, continuous use	Suits short-term or specialized needs

Buy	Lease
Risk of obsolescence	Lessor bears obsolescence (operating lease)

Safety in projects: hazard identification and risk assessment (HIRA), personal protective equipment, permits to work, lockout-tagout, scaffolding and work at height rules, fire safety, training, toolbox talks, accident reporting. Laws: Factories Act, Building and Other Construction Workers Act, OSH Code 2020.

Environmental factors: EIA, consent to establish and operate from pollution control boards, dust, noise, waste and water management, tree replanting, environmental management plan.

Sources of project finance: equity (promoters, IPO, venture capital), debt (term loans, bonds, debentures), external commercial borrowing, government grants and subsidies, viability gap funding, lease finance, supplier credit, PPP models (BOT, BOOT, BOO, DBFO, HAM). Financial appraisal uses NPV, IRR, DSCR (debt service coverage ratio = cash available for debt service / debt service).

International projects: issues: currency risk, political and legal risk, cultural differences, language, time zones, local regulations and taxes, logistics, standards. Tools: hedging, political risk insurance, local partners, cross-cultural training.

Joint ventures: types: equity JV (new company) and contractual JV (consortium). Reasons: combine skills, share risk and capital, enter local markets, meet local content rules. Issues: control, profit sharing, culture clash, exit terms; needs a clear JV agreement.

Cultural impact (Hofstede's dimensions): power distance, individualism vs collectivism, masculinity vs femininity, uncertainty avoidance, long-term orientation, indulgence. Project managers must adapt communication, decision making and motivation to the local culture.

Handover: commissioning and testing, punch list (remaining minor defects) cleared, as-built drawings, operation and maintenance manuals, spares, warranties and guarantees, training of operators, handover certificate, defect liability period.

🧮 Solved examples

Example 1: Site selection by factor rating.

Factor	Weight	Site P	Site Q
Land cost	0.25	7	9
Access to market	0.30	9	6
Labour availability	0.20	8	7
Power and water	0.25	6	8

- P = 0.25×7 + 0.30×9 + 0.20×8 + 0.25×6 = 1.75 + 2.7 + 1.6 + 1.5 = **7.55**.
- Q = 0.25×9 + 0.30×6 + 0.20×7 + 0.25×8 = 2.25 + 1.8 + 1.4 + 2.0 = **7.45**. Choose site P.

Example 2: Buy vs lease (simple). A crane costs ₹40 lakh and can be sold for ₹10 lakh after 3 years; operating cost is the same either way. Leasing costs ₹12 lakh per year. Ignoring interest, buying costs 40 − 10 = ₹30 lakh, leasing costs 36 lakh, so buying is cheaper. If the crane is needed for only 1 year, leasing costs ₹12 lakh against a large loss on early resale, so leasing is better.

Example 3: DSCR. Cash available for debt service ₹30 crore; annual principal plus interest ₹20 crore. DSCR = 30/20 = 1.5, above the usual lender minimum of about 1.2–1.3, so the project can service its debt.

Key points to remember

- Site selection factors; preparation steps; layout principles.
- Equipment selection factors; buy vs lease table.
- Safety: HIRA, PPE, permits, LOTO; environment: EIA, consents, EMP.
- Finance sources; PPP models (BOT, BOOT, HAM); DSCR.
- International projects: currency, political, cultural risks; JV types.
- Handover: commissioning, punch list, as-builts, O&M manuals, training, defect liability.

⭐ Important questions

1. Explain site mapping, site suitability, site layout and site preparation.
2. Explain equipment selection and the buy vs lease decision.
3. Explain safety and environmental factors in projects.
4. Explain the sources of project finance and PPP models.
5. Explain the management of international projects, joint ventures and cultural issues.
6. Explain the handover of a project.

Last-hour checklist

- ☐ Project definition and characteristics; project vs operations; types.
- ☐ PMBOK 10 areas and 5 process groups; PM role; organization structures table.
- ☐ Life cycle; charter, sponsor; planning steps; feasibility types; SWOT; RACI; Tuckman.
- ☐ Selection methods; BEP (5,000 units); NPV (₹1,37,236), payback (3.33 years), PI.
- ☐ Cost estimation methods; cost vs price vs value.
- ☐ CPM forward and backward pass; floats; critical path A-D-F (13 weeks).
- ☐ PERT t_e and variance; probability 0.8413; CPM vs PERT.
- ☐ Crashing (₹3,500 to reach 11 weeks); cost slope; decision tree (EMV 38 vs 32); risk responses.
- ☐ Vendor rating; JIT; supply chain; quality tools; QA vs QC; ISO 9001; TQM; Six Sigma.
- ☐ Resource smoothing vs levelling.
- ☐ Change control; EVM formulas and example (CPI 0.8, EAC ₹12.5 lakh); audit; close-out types.
- ☐ Shutdown and turnaround; replacement (replace after 5 years).
- ☐ Site factors and preparation; buy vs lease; safety and EIA; finance and PPP; international projects and JVs; handover.

Books prescribed by AKTU

These are the books named in the official AKTU syllabus. Buy or borrow them through a bookshop or your college library. The WorldCat links show which libraries hold a copy.

#	Title	Author(s)	Publisher	Where to get it legally
1	Essentials of Project Management	Kamaraj Ramakrishna	PHI Learning (2010)	WorldCat
2	Projects: Planning, Analysis, Selection, Financing, Implementation and Review	Prasanna Chandra	Tata McGraw-Hill (2010)	WorldCat
3	Construction Project Management: Planning, Scheduling and Controlling	K. K. Chitkara	Tata McGraw-Hill	WorldCat
4	Project Management: A Systems Approach to Planning, Scheduling, and Controlling	Harold Kerzner	Wiley	WorldCat

The syllabus copy also lists two electric-vehicle books under this course, which looks like a printing error, so they are left out here.

Syllabus source: AKTU Open Elective syllabus for B.Tech final year, 2025-26 (copy hosted by an affiliated college, PDF)

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